The results were encouraging. Not only did the researchers find that the chances of producing strange matter or black holes were incredibly small based just on theoretical considerations, there actually was experimental data to back this up. Specifically: the existence of the Moon.

The argument for any kind of weird collider-induced phenomenon destroying us rests on the notion that the extreme high-energy collisions in these colliders are so unprecedented that we couldn't possibly know what might happen. Which is ignoring an important fact: while the energies reached by RHIC and LHC might be novel to us puny humans, cosmic rays cruising through the universe reach incredibly high energies all the time, and collide with other objects and each other constantly. In the words of the RHIC paper authors, "It is clear that cosmic rays have been carrying out RHIC-like 'experiments' throughout the universe since time out of mind." Collisions at much higher energies than what any Earthly collider could reach have been occurring across the universe for billions of years, so if they could destroy the cosmos, surely we would have noticed.

"Hang on," you might say, "what if cosmic ray collisions in deep space really are incredibly destructive, but just too far away to affect us? What if strange matter clumps exist all over the cosmos, and we just don't know?" It's a valid concern. While most of the time, particles produced in a collider are expected to have enough leftover momentum to zip out of the lab as soon as they're formed, it's conceivable that we could create something dangerous that could more or less come to rest in the detector. What then?

Fortunately, we can use the Moon as our canary in the coal mine. We have enough data from Earth-based detectors and space telescopes to know that high-energy cosmic rays slam into the Moon *all the time*. (In fact, with radio telescopes we can even use the Moon as a neutrino detector,[fn4] which is kind of amazing in and of itself.) If high-energy particle collisions could convert nearby ordinary matter into strange matter, this would have happened on the Moon eons ago, and we would have a VERY different object in our sky. Likewise, the night sky would be pretty noticeably altered if a tiny black hole formed on the Moon and swallowed it up. Not to mention the fact that we humans have actually *been there*, walked around, hit a few golf balls, and brought back samples. The Moon is doing just fine. Ergo, the authors argued, the RHIC won't kill us all.

Strange matter and black holes weren't the only apocalypses debunked, though. Another possibility, similarly dismissed by witnessing the superior